

SUPERMARKET SALES ANALYSIS

Data Analytics Project

IBM SkillsBuild Data Analytics with AI Academic Internship Program 2026

Project Report

1. Introduction

This project analyzes supermarket sales transactions to identify useful patterns in sales, products, branches, customer types, payment methods, quantities and customer ratings. The purpose is to demonstrate a practical data analytics workflow using Python, Pandas, NumPy, Matplotlib and Jupyter Notebook.

2. Problem Statement

The objective is to transform raw supermarket transaction data into meaningful business insights. The analysis evaluates product and category performance, branch performance, payment preferences, customer spending behavior and customer satisfaction.

3. Dataset

The project uses a supermarket sales dataset containing 500 sales transactions. The dataset includes information such as product/category, branch, city, customer type, quantity, unit price, payment method, sales and customer rating.

Dataset link:

https://docs.google.com/spreadsheets/d/1QIX__4VObHFMEXnRM2xJyXmB5JAB2peHrJcQ41_U9TE/edit?usp=sharing

4. Tools and Technologies

- Python
- Pandas

- NumPy
- Matplotlib
- Jupyter Notebook
- Excel

5. Data Cleaning and Validation

- Loaded the Excel dataset into a Pandas DataFrame.
- Inspected the dataset shape, columns and data types.
- Checked for missing values.
- Checked for duplicate records.
- Converted relevant fields to numeric data types.
- Validated the sales calculation using $\text{Quantity} \times \text{Unit Price}$.
- Checked customer ratings for values outside the expected 0–5 range.

6. Exploratory Data Analysis

Exploratory analysis was performed by grouping and aggregating the dataset across products/product lines, branches, cities, payment methods and customer types. Matplotlib visualizations were used to compare sales and transaction counts.

7. KPI Calculations

KPI	Result
Total Transactions	500
Total Sales	₹244,411.08
Highest-selling product	Cheese — ₹27,906.30
Best-performing branch	Branch C (Mumbai) — ₹72,469.45
Highest-selling category	Beverages — ₹56,108.24
Most-used payment method	UPI — 127 transactions
Average Member transaction	₹483.14
Average Normal transaction	₹497.07
Average customer rating	3.99 / 5

8. Business Questions and Findings

Which product generates the highest sales? Cheese generated the highest sales at ₹27,906.30.

Which branch performs best? Branch C in Mumbai generated the highest branch sales at ₹72,469.45.

Which category sells the most? Beverages generated the highest category sales at ₹56,108.24.

What is the most popular payment method? UPI was the most-used payment method with 127 transactions.

Do Members spend more than Normal customers? No. The average Member transaction was ₹483.14, while the Normal customer average was ₹497.07.

What is the average customer rating? The average customer rating was 3.99 out of 5.

9. Business Recommendations

1. Maintain sufficient inventory for high-selling products and categories.
2. Study the operational factors associated with strong performance at Branch C.
3. Continue supporting UPI and other commonly used payment methods.
4. Use customer ratings to identify service-quality improvement opportunities.
5. Compare Member and Normal purchasing patterns before designing membership offers.
6. Use sales and product trends to improve inventory allocation and promotional planning.

10. Conclusion

The project demonstrates how raw supermarket transaction data can be cleaned, analyzed and visualized to produce practical business insights. Python-based exploratory analysis makes it possible to understand sales

performance, customer behavior, payment preferences and customer satisfaction. These findings can support inventory planning, branch performance analysis, customer-service improvement and business decision-making.

11. Project Files

- DeepakMalviya_SupermarketSalesAnalysis.ipynb — complete project code
- requirements.txt — Python dependencies
- README.md — project overview and setup instructions
- DeepakMalviya_ProjectReport.docx — this project report